SOLID PLASTIC AND E-WASTE RULES

THREE-STREAM FIREWALL → RULE-VINTAGE LADDER → MUNICIPAL RESPONSIBILITY MAP → SEGREGATION-TO-DISPOSAL CHAIN → PLASTIC ACTOR MATRIX → BAN-VERSUS-EPR GATE

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ● SUBORDINATE EXTRA ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g2 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 THREE-STREAM FIREWALL

THREE-STREAM FIREWALL | SOLID WASTE | MUNICIPAL GENERATION, SEGREGATION AND LOCAL SERVICES | PLASTIC WASTE | PRODUCT AND PACKAGING CONTROLS PLUS EPR | ELECTRICAL AND ELECTRONIC END-OF-LIFE CHAIN | COMMON PARENT | ENVIRONMENT PROTECTION ACT FRAMEWORK

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
SOLID WASTE → municipal generation, segregation and local services	E-WASTE → electrical and electronic end-of-life chain	RULE → separate streams remain legally distinct
PLASTIC WASTE → product and packaging controls plus EPR	COMMON PARENT → Environment Protection Act framework	SOLID WASTE → municipal generation, segregation and local services

ECOLOGICAL UNIT / PARAMETER
- SOLID WASTE → municipal generation, segregation and local services
- PLASTIC WASTE → product and packaging controls plus EPR

DECISIVE DISTINCTION
- E-WASTE → electrical and electronic end-of-life chain
- COMMON PARENT → Environment Protection Act framework

SCALE / STATUS / EXCEPTION
- RULE → separate streams remain legally distinct
- SOLID WASTE → municipal generation, segregation and local services

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PLASTIC WASTE → product and packaging controls plus EPR.

01

STAGE 01 RULE-VINTAGE LADDER

RULE-VINTAGE LADDER | BASE RULE | IDENTIFY NOTIFICATION | IDENTIFY CHANGED PROVISION AND DATE | GUIDELINE OR SOP | IMPLEMENTATION INSTRUCTION | PORTAL PROCEDURE | DIGITAL COMPLIANCE WORKFLOW

BASE RULE → identify notification → AMENDMENT → identify changed provision and date → GUIDELINE OR SOP → implementation instruction → PORTAL PROCEDURE → digital compliance workflow → CURRENT CLAIM → verify all later changes

SYSTEM / LEVEL / DEFINITION
- BASE RULE → identify notification
- AMENDMENT → identify changed provision and date

STRUCTURE-FUNCTION MECHANISM
- GUIDELINE OR SOP → implementation instruction
- PORTAL PROCEDURE → digital compliance workflow

SCALE / EVIDENCE LIMIT
- CURRENT CLAIM → verify all later changes
- BASE RULE → identify notification

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PORTAL PROCEDURE → digital compliance workflow; CURRENT CLAIM → verify all later changes.

02

STAGE 02 MUNICIPAL RESPONSIBILITY MAP

MUNICIPAL RESPONSIBILITY MAP | SEGREGATE AND HAND OVER | BULK GENERATOR | ADDITIONAL APPLICABLE DUTIES | COLLECTION AND PROCESSING SYSTEM | RUN AUTHORISED FACILITY | SPCB OR PCC | AUTHORISATION AND MONITORING

GENERATOR → segregate and hand over

BULK GENERATOR → additional applicable duties → ULB → collection and processing system → OPERATOR → run authorised facility

SYSTEM / LEVEL / DEFINITION
- GENERATOR → segregate and hand over
- BULK GENERATOR → additional applicable duties

STRUCTURE-FUNCTION MECHANISM
- ULB → collection and processing system
- OPERATOR → run authorised facility

SCALE / EVIDENCE LIMIT
- SPCB OR PCC → authorisation and monitoring
- GENERATOR → segregate and hand over

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OPERATOR → run authorised facility; SPCB OR PCC → authorisation and monitoring.

03

STAGE 03 SEGREGATION-TO-DISPOSAL CHAIN

SEGREGATION-TO-DISPOSAL CHAIN | PREVENT OR REDUCE | AVOID WASTE | PRESERVE RECOVERABLE STREAMS | COLLECT AND TRANSPORT | MOVE WITHOUT REMIXING | RECOVER OR TREAT | PROCESS BY STREAM

PREVENT OR REDUCE → avoid waste → SEGREGATE → preserve recoverable streams → COLLECT AND TRANSPORT → move without remixing → RECOVER OR TREAT → process by stream → RESIDUE → safe final disposal

SYSTEM INPUT / STARTING CONDITION
- PREVENT OR REDUCE → avoid waste

ECOLOGICAL PROCESS / TRANSFER
- COLLECT AND TRANSPORT → move without remixing

OUTCOME / FEEDBACK / LIMIT
- RESIDUE → safe final disposal

- GENERATOR → segregate and hand over
- BULK GENERATOR → additional applicable duties

- ULB → collection and processing system
- OPERATOR → run authorised facility

- SPCB OR PCC → authorisation and monitoring
- GENERATOR → segregate and hand over

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OPERATOR → run authorised facility; SPCB OR PCC → authorisation and monitoring.

03

STAGE 03

SEGREGATION-TO-DISPOSAL CHAIN

SEGREGATION-TO-DISPOSAL CHAIN · PREVENT OR REDUCE · AVOID WASTE · PRESERVE RECOVERABLE STREAMS · COLLECT AND TRANSPORT · MOVE WITHOUT REMIXING · RECOVER OR TREAT · PROCESS BY STREAM

PREVENT OR REDUCE → avoid waste → **SEGREGATE → preserve recoverable streams** → **COLLECT AND TRANSPORT → move without remixing** → **RECOVER OR TREAT → process by stream** → **RESIDUE → safe final disposal**

SYSTEM INPUT / STARTING CONDITION
- PREVENT OR REDUCE → avoid waste
- SEGREGATE → preserve recoverable streams

ECOLOGICAL PROCESS / TRANSFER
- COLLECT AND TRANSPORT → move without remixing
- RECOVER OR TREAT → process by stream

OUTCOME / FEEDBACK / LIMIT
- RESIDUE → safe final disposal
- PREVENT OR REDUCE → avoid waste

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RECOVER OR TREAT → process by stream; RESIDUE → safe final disposal.

04

STAGE 04

PLASTIC ACTOR MATRIX

PLASTIC ACTOR MATRIX · COVERED PRODUCT OR PACKAGING RESPONSIBILITY · COVERED IMPORTED PRODUCT RESPONSIBILITY · BRAND OWNER · BRANDED PACKAGING RESPONSIBILITY · AUTHORISED END-OF-LIFE ACTIVITY · MUNICIPAL SERVICE ROLE, NOT AUTOMATIC EPR SUBSTITUTE

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
PRODUCER → covered product or packaging responsibility	BRAND OWNER → branded packaging responsibility	ULB → municipal service role, not automatic EPR substitute
IMPORTER → covered imported product responsibility	PROCESSOR → authorised end-of-life activity	PRODUCER → covered product or packaging responsibility

ECOLOGICAL UNIT / PARAMETER
- PRODUCER → covered product or packaging responsibility
- IMPORTER → covered imported product responsibility

DECISIVE DISTINCTION
- BRAND OWNER → branded packaging responsibility
- PROCESSOR → authorised end-of-life activity

SCALE / STATUS / EXCEPTION
- ULB → municipal service role, not automatic EPR substitute
- PRODUCER → covered product or packaging responsibility

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ULB → municipal service role, not automatic EPR substitute.

05

STAGE 05

BAN-VERSUS-EPR GATE

BAN-VERSUS-EPR GATE · IDENTIFIED ITEM · TEST THE EXACT RESTRICTION AND VINTAGE · PLASTIC PACKAGING · TEST EPR CATEGORY AND OBLIGATION · OTHER PRODUCT · DO NOT INFER BAN FROM PLASTIC CONTENT · MATERIAL FACT

IDENTIFIED ITEM → test the exact restriction and vintage → **PLASTIC PACKAGING → test EPR category and obligation** → **OTHER PRODUCT → do not infer ban from plastic content** → **MATERIAL FACT → composition question** → **LEGAL FACT → rule-specific classification**

SYSTEM / LEVEL / DEFINITION
- IDENTIFIED ITEM → test the exact restriction and vintage
- PLASTIC PACKAGING → test EPR category and obligation

STRUCTURE-FUNCTION MECHANISM
- OTHER PRODUCT → do not infer ban from plastic content
- MATERIAL FACT → composition question

SCALE / EVIDENCE LIMIT
- LEGAL FACT → rule-specific classification
- IDENTIFIED ITEM → test the exact restriction and vintage

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OTHER PRODUCT → do not infer ban from plastic content.

06

STAGE 06

EPR EVIDENCE LADDER

EPR EVIDENCE LADDER · OBLIGATION OR TARGET · LEGAL REQUIREMENT · ACTOR ENTERS PORTAL · COMPLIANCE INSTRUMENT · REPORTED THROUGHPUT · CLAIMED PROCESSING · VERIFIED RECOVERY

OBLIGATION OR TARGET → legal requirement

REGISTRATION → actor enters portal → **CERTIFICATE → compliance instrument** → **REPORTED THROUGHPUT → claimed processing**

SYSTEM / LEVEL / DEFINITION
- OBLIGATION OR TARGET → legal requirement
- REGISTRATION → actor enters portal

STRUCTURE-FUNCTION MECHANISM
- CERTIFICATE → compliance instrument
- REPORTED THROUGHPUT → claimed processing

SCALE / EVIDENCE LIMIT
- VERIFIED RECOVERY → physical outcome evidence
- OBLIGATION OR TARGET → legal requirement

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OTHER PRODUCT → do not infer ban from plastic content.

06

STAGE 06 EPR EVIDENCE LADDER

EPR EVIDENCE LADDER · OBLIGATION OR TARGET · LEGAL REQUIREMENT · ACTOR ENTERS PORTAL · COMPLIANCE INSTRUMENT · REPORTED THROUGHPUT · CLAIMED PROCESSING · VERIFIED RECOVERY

OBLIGATION OR TARGET → legal requirement

REGISTRATION → actor enters portal → CERTIFICATE → compliance instrument → REPORTED THROUGHPUT → claimed processing

SYSTEM / LEVEL / DEFINITION
- OBLIGATION OR TARGET → legal requirement
- REGISTRATION → actor enters portal

STRUCTURE-FUNCTION MECHANISM
- CERTIFICATE → compliance instrument
- REPORTED THROUGHPUT → claimed processing

SCALE / EVIDENCE LIMIT
- VERIFIED RECOVERY → physical outcome evidence
- OBLIGATION OR TARGET → legal requirement

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: REPORTED THROUGHPUT → claimed processing; VERIFIED RECOVERY → physical outcome evidence.

07

STAGE 07 PROCESSOR AUDIT GATE

PROCESSOR AUDIT GATE · FORMAL ELIGIBILITY · INPUT RECORD · MATERIAL RECEIVED · PROCESS RECORD · ACTUAL AUTHORISED OPERATION · OUTPUT AND RESIDUE · DESTINATION AND SAFE HANDLING

REGISTRATION → formal eligibility

INPUT RECORD → material received → PROCESS RECORD → actual authorised operation → OUTPUT AND RESIDUE → destination and safe handling

SYSTEM INPUT / STARTING CONDITION
- REGISTRATION → formal eligibility
- INPUT RECORD → material received

ECOLOGICAL PROCESS / TRANSFER
- PROCESS RECORD → actual authorised operation
- OUTPUT AND RESIDUE → destination and safe handling

OUTCOME / FEEDBACK / LIMIT
- AUDIT → quantity and outcome require verification
- REGISTRATION → formal eligibility

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OUTPUT AND RESIDUE → destination and safe handling.

08

STAGE 08 E-WASTE ACTOR CHAIN

E-WASTE ACTOR CHAIN · MANUFACTURER OR IMPORTER · PRODUCT ENTERS MARKET · EPR-SIDE RESPONSIBILITY · CONSUMER OR BULK CONSUMER · CHANNELISATION DUTY · LIFE EXTENSION WHERE APPLICABLE · AUTHORISED MATERIAL RECOVERY

MANUFACTURER OR IMPORTER → product enters market → PRODUCER → EPR-side responsibility → CONSUMER OR BULK CONSUMER → channelisation duty → REFURBISHER → life extension where applicable → RECYCLER → authorised material recovery

SYSTEM INPUT / STARTING CONDITION
- MANUFACTURER OR IMPORTER → product enters market
- PRODUCER → EPR-side responsibility

ECOLOGICAL PROCESS / TRANSFER
- CONSUMER OR BULK CONSUMER → channelisation duty
- REFURBISHER → life extension where applicable

OUTCOME / FEEDBACK / LIMIT
- RECYCLER → authorised material recovery
- MANUFACTURER OR IMPORTER → product enters market

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: REFURBISHER → life extension where applicable; RECYCLER → authorised material recovery.

09

STAGE 09 FORMAL-INFORMAL INTEGRATION MAP

FORMAL-INFORMAL INTEGRATION MAP · INFORMAL COLLECTOR · REACH AND LIVELIHOOD · UNSAFE DISMANTLING · EXPOSURE AND POLLUTION RISK · AUTHORISED FACILITY · CONTROLLED PROCESSING ROUTE · RECOGNITION, SAFETY AND TRACEABILITY

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
INFORMAL COLLECTOR → reach and livelihood	AUTHORISED FACILITY → controlled processing route	VERDICT → formalise processing without erasing livelihoods
UNSAFE DISMANTLING → exposure and pollution risk	INTEGRATION → recognition, safety and traceability	INFORMAL COLLECTOR → reach and livelihood

ECOLOGICAL UNIT / PARAMETER
- INFORMAL COLLECTOR → reach and livelihood
- UNSAFE DISMANTLING → exposure and pollution risk

DECISIVE DISTINCTION
- AUTHORISED FACILITY → controlled processing route
- INTEGRATION → recognition, safety and traceability

SCALE / STATUS / EXCEPTION
- VERDICT → formalise processing without erasing livelihoods
- INFORMAL COLLECTOR → reach and livelihood

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INTEGRATION → recognition, safety and traceability; VERDICT → formalise processing without erasing livelihoods.

09

STAGE 09 FORMAL-INFORMAL INTEGRATION MAP

FORMAL-INFORMAL INTEGRATION MAP · INFORMAL COLLECTOR · REACH AND LIVELIHOOD · UNSAFE DISMANTLING · EXPOSURE AND POLLUTION RISK · AUTHORISED FACILITY · CONTROLLED PROCESSING ROUTE · RECOGNITION, SAFETY AND TRACEABILITY

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
INFORMAL COLLECTOR → reach and livelihood	AUTHORISED FACILITY → controlled processing route	VERDICT → formalise processing without erasing livelihoods
UNSAFE DISMANTLING → exposure and pollution risk	INTEGRATION → recognition, safety and traceability	INFORMAL COLLECTOR → reach and livelihood

ECOLOGICAL UNIT / PARAMETER

- INFORMAL COLLECTOR → reach and livelihood
- UNSAFE DISMANTLING → exposure and pollution risk

DECISIVE DISTINCTION

- AUTHORISED FACILITY → controlled processing route
- INTEGRATION → recognition, safety and traceability

SCALE / STATUS / EXCEPTION

- VERDICT → formalise processing without erasing livelihoods
- INFORMAL COLLECTOR → reach and livelihood

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INTEGRATION → recognition, safety and traceability; VERDICT → formalise processing without erasing livelihoods.

10

STAGE 10 STOCK-FLOW AND TECHNOLOGY MATRIX

STOCK-FLOW AND TECHNOLOGY MATRIX · CURRENT WASTE FLOW · EPR AND COLLECTION CONTROLS · LEGACY DUMP OR SITE · REMEDIATION PROBLEM · MATERIAL RECOVERY ROUTE · THERMAL OR CHEMICAL ROUTE · FEEDSTOCK AND EMISSION CONDITIONS

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
CURRENT WASTE FLOW → EPR and collection controls	RECYCLING → material recovery route	NO UNIVERSAL WINNER → compare complete mass and residue balance
LEGACY DUMP OR SITE → remediation problem	THERMAL OR CHEMICAL ROUTE → feedstock and emission conditions	CURRENT WASTE FLOW → EPR and collection controls

ECOLOGICAL UNIT / PARAMETER

- CURRENT WASTE FLOW → EPR and collection controls
- LEGACY DUMP OR SITE → remediation problem

DECISIVE DISTINCTION

- RECYCLING → material recovery route
- THERMAL OR CHEMICAL ROUTE → feedstock and emission conditions

SCALE / STATUS / EXCEPTION

- NO UNIVERSAL WINNER → compare complete mass and residue balance
- CURRENT WASTE FLOW → EPR and collection controls

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NO UNIVERSAL WINNER → compare complete mass and residue balance.

11

STAGE 11 WASTE-GOVERNANCE ANSWER SPINE

WASTE-GOVERNANCE ANSWER SPINE · STREAM, MATERIAL AND RULE VINTAGE · GENERATOR, ULB, PRODUCER, IMPORTER AND PROCESSOR · SEGREGATION, COLLECTION, PROCESSING AND RESIDUE · OBLIGATION, CERTIFICATE, THROUGHPUT AND RECOVERY · INFORMAL SECTOR, LEGACY STOCK, TECHNOLOGY AND PYQ LIMITS

CLASSIFY → stream, material and rule vintage → ASSIGN → generator, ULB, producer, importer and processor → TRACE → segregation, collection, processing and residue → VERIFY → obligation, certificate, throughput and recovery → QUALIFY → informal sector, legacy stock, technology and PYQ limits

SYSTEM / LEVEL / DEFINITION

- CLASSIFY → stream, material and rule vintage
- ASSIGN → generator, ULB, producer, importer and processor

STRUCTURE-FUNCTION MECHANISM

- TRACE → segregation, collection, processing and residue
- VERIFY → obligation, certificate, throughput and recovery

SCALE / EVIDENCE LIMIT

- QUALIFY → informal sector, legacy stock, technology and PYQ limits
- CLASSIFY → stream, material and rule vintage

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: QUALIFY → informal sector, legacy stock, technology and PYQ limits.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · INDIA'S INFORMAL WASTE ECONOMY (WASTE-PICKERS, KABADIWALAS/SCRAP DEALERS, INFORMAL · POLICY TENSION · FORMAL EPR SYSTEMS RISK BYPASSING OR DISPLACING THIS INFORMAL · THIS IS A RECURRING, GENUINELY TWO-SIDED POLICY DILEMMA · PURELY INFORMAL PROCESSING · INDIA'S SINGLE-USE-PLASTIC RESTRICTIONS TARGET SPECIFICALLY IDENTIFIED ITEMS (BASED ON · ENFORCEMENT REALITY

OPTIONAL ECOLOGICAL PROCESS DEPTH

- CAUTION: India's informal waste economy (waste-pickers, kabadiwalas/scrap dealers, informal
- CAUTION: Policy tension: formal EPR systems risk bypassing or displacing this informal
- CAUTION: This is a recurring, genuinely two-sided policy dilemma: purely informal processing
- India's single-use-plastic restrictions target specifically identified items (based on

SCALE, PARAMETER OR STATUS LIMIT

- CAUTION: Enforcement reality: effective enforcement of item-specific bans requires
- CAUTION: National aggregate figures for plastic-waste generation, collection rate and recycling
- CAUTION: E-waste generation growth estimates are typically modelled from electronics-sales data
- CPCB's EPR portals allow producers to meet obligations by acquiring EPR certificates

COMPARATIVE RESTORATION NUANCE

- India's informal waste sector has historically handled a substantial share of actual
- India's single-use-plastic restrictions target specifically identified items rather
- National plastic/e-waste generation and recycling-rate figures are largely modelled/
- NOT: EPR compliance requires a producer's own waste to be physically recycled by a specific

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels